

# Under five minutes, **ten slides.**

What to say on every slide, **29 September.**

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## Run of show · 29 Sept

Under five minutes, ten slides: 460 spoken words and 15 clicks, 3:38 at 150 words a minute. Detail that is not on a slide is in the notes here. The spine: **shown, felt, lived** (02, 03, 04), then **the map, why it compounds, the product** (05, 06, 07), then **who else, the money, the ask** (08, 09, 10). 08 is the fastest slide: one beat.

| SLIDE                       | CLOCK     | SEC | CLICKS | BEAT         | ONE LINE TO LAND                                               |
|-----------------------------|-----------|-----|--------|--------------|----------------------------------------------------------------|
| 01 Do my cells look normal? | 0:00–0:21 | 21  | 1      | hook         | Drop an image. Get an answer against every other lab.          |
| 02 One lab, or six?         | 0:21–0:54 | 33  | 3      | shown        | The model isn't the hard part. The map is.                     |
| 03 Do my cells look normal? | 0:54–1:21 | 27  | 1      | felt         | Every company that buys cell data gets this problem delivered. |
| 04 Founder                  | 1:21–1:46 | 25  | 2      | lived        | This is the thing I kept needing and nobody sold.              |
| 05 The map                  | 1:46–2:04 | 18  | 2      | the map      | Where you sit predicts what you measure.                       |
| 06 Open at the core         | 2:04–2:28 | 25  | 1      | the standard | The map is open. Access to it structured is what sells.        |

| SLIDE                               | CLOCK         | SEC | CLICKS | BEAT        | ONE LINE TO LAND                             |
|-------------------------------------|---------------|-----|--------|-------------|----------------------------------------------|
| 07 The product                      | 2:28–<br>3:04 | 35  | 2      | what        | Send your controls. See where you stand.     |
| 08 Five questions, five tools       | 3:04–<br>3:22 | 18  | 1      | who else    | Use all of them. The last one did not exist. |
| 09 Eighteen months, four milestones | 3:22–<br>3:34 | 12  | 1      | the money   | A first paid certificate.                    |
| 10 Every plate, placed on the map   | 3:34–<br>3:59 | 25  | 0      | vision, ask | Have images of U2OS cells?<br>Talk to me.    |

Captions stay hidden while you present (Tweaks › captions); the shared link and PDF show them. If you're late, skip the second sentence of 03 and the first sentence of 10.

## 01 Do my cells look normal? (0:00–0:21)

*On screen while you're introduced. The box is the product; the dial is the map. Say the question, then the sentence.*

**0:00** Hi, I'm Kat. Every result in biology is a difference from a control, and nobody checks the control. Datum Line is a box: drop a control image, get an answer against every other lab. Labs ask for free. The companies that buy cell data pay for the certificate. *Click.*

**Precision to hold.** Today the box takes a control-well profile from any lab in the JUMP format; a raw image from any microscope is the bet, and the test of it is results/e15 (board A19 when it exists). Say "drop an image" and, if asked, say exactly that.

## 02 One lab, or six? (0:21–0:54, three views)

*Shown. The first view is a game: let the room look for four seconds.*

**0:21** Six untreated wells. Same line, same protocol, no drug. One lab, or six? *Click.*

**0:28** Six. Nobody here can tell. A classifier can: 99.9 percent across eleven labs, against nine by chance, on plates it never saw. *Click.*

**0:39** Why now: this data is public and free, and models are a commodity. The model isn't the hard part. The map is. *Click.*

**Precision to hold.** Eleven labs, 93,228 untreated wells, 1,872 plates, held out by plate; 78 wells wrong. Holding out whole batches gives 99.6%; labs on one microscope model, 99.99%. The signature is in every feature family, image-quality features included, so it reflects the whole imaging pipeline as well as the cells. Say "the lab", not "the cells". Cell Painting fluorescence; brightfield is the bet.

**For ML people.** The line on the second view: normalising each plate to its own controls takes lab identity from 99.9 to 24 percent. Right for discovery, wrong for QC (A3).

**Why now, in full.** Three things that were not true three years ago. JUMP made the field's reference public and CC0 in 2023 [7], so the map exists and it works on eleven labs. Image models are a commodity, so nobody wins on the model; they win on the reference. And on 10 April 2025 the FDA announced a plan to phase out animal-testing requirements in favour of new approach methods [9], so cell-based data starts to carry regulatory weight, and data that carries regulatory weight needs a calibration standard. Nobody holds it yet.

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### **03 Do my cells look normal? (0:54–1:21)**

*Felt. Point at the pink slice, then the six search results, then the two edges of card 3: run it, or hold.*

**0:54** So every lab asks the same question, and can't answer it. About twenty-eight billion dollars a year of US preclinical research can't be reproduced, and a third of that is inputs.

**1:06** Today you search, you ask around, then you guess. I've seen one lab run too fast and one hold too long. And every company that buys cell data gets this problem delivered. *Click.*

**Precision to hold.** \$28B is a 2015 US estimate; the 36% covers all reagents and reference materials, not only cells. The search card shows six real U2OS images from six papers, 2009 to 2023, via IDR and JUMP; sources are in the README. Card 3 names neither lab and gives no exact figure; keep it that way on stage. Growth chart is the metaphor if you need one: nobody plots their cells against the population.

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#### **04 Founder (1:21–1:46)**

*Lived. Let the logos and the pink line under your name do the credential work.*

**1:21** I've asked it at five benches. Nobody could answer it. *Click.*

**1:27** At Axiom we onboarded hepatocytes through a good CRO. Their checks passed; ours passed. The first sign anything had moved came from outside: the Broad couldn't align our profiles to their reference.

**1:40** This is the thing I kept needing and nobody sold. *Click.*

**Founder detail, for the judge question (about 30 seconds).** Three methods, three papers, one thread. At AbbVie I built a neuronal tau aggregation assay for drug discovery [3]: the readout was what the cells did, not a label on them. At Axiom I built Cell Painting in primary human hepatocytes for cytotoxicity and mode of action [1], ran assay development for two of the largest Cell Painting datasets in the world, and invented two bioactivity detection methods that see a small molecule's effect on cells at about ten times lower concentration than LDH, ATP or cell count. With the Carpenter lab we showed that cell count alone predicts small-molecule bioactivity benchmarks [2]: even the simplest phenotype carries the signal. The thread is that bioactivity is a measurement of behaviour, and behaviour is the half of QC nobody measures. Every lab already images its control wells; Datum Line turns that image into the certificate, against the field.

**Precision to hold.** The question predates plates: at MGH it was whether a *Xenopus* oocyte, a single frog egg cell, looked normal before a recording. Say only what your author contribution statements support: "built" on [1] and [3], "we showed" on [2]. The two bioactivity methods and the ten-fold sensitivity are from your resume (macrofluidic.me/resume); if asked, name the comparison assays and the concentration range. The 2019 GABA<sub>A</sub> paper [5] is the MGH anesthesia lab, Harvard Medical School, 2017 to 2018. The industry claim is the deck's own: STR and vendor certificates cover identity, bioactivity covers behaviour, and behaviour is what a control well measures in any lab that images one.

**Precision to hold.** "Published with" lists coauthor institutions, never partners. The Axiom story has no villain: the CRO passed every check it had, Axiom passed every check it had, and the failure was that no check for this existed. Name no vendor and no person. The uneven distribution was noticed; its severity and its effect on data quality were unknown, so never present it as the cause, and never say "spun down unevenly". If asked what it was: "We still don't know for certain. The uneven spreading is one hypothesis; whole-well images show it more easily than our fields, and a field reference is what would tell you whether it matters." The sequence is: local checks passed, uneven spreading noticed but unquantified, the Broad flagged that reference profiles would not align, the cause is still open. Say what happened, not how it was fixed. Confirm with Axiom that you can tell it; if not, say "at one lab".

**Why me, in one breath.** I have been on every side of the transaction: I built the assays (AbbVie, insitro), I ran the outsourced data purchase that did not align (Axiom, the CRO, the Broad flagging it), I invented bioactivity methods that measure what cells do rather than what the label says, and I published with the Broad, which built the reference. The buyer, the maker and the method inventor in one person is the honest reason founder comes before product.

**Team, honestly.** One founder today, and say so before anyone asks. I am looking for a technical co-founder, and I am in conversation with one of the people I wrote the cell-counting paper with [2]; no name on a public page until it is agreed. The other

collaborators are in industry, as partners, not hires. If asked why not my Axiom colleagues: they are building Axiom, and I will not hire from a company I helped build.

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## 05 The map (1:46–2:04, two views)

*The map. Dark slide: slow down. Two views, one sentence each.*

**1:46** Eleven labs, same cells, same protocol: eleven different normals. That's what the classifier saw. *Click.*

**1:53** And where a lab sits on this map predicts how far its drug answers have moved. The map is worth having. *Click.*

**Figure detail.** First view: a PCA of 15,352 untreated wells for the eye; distances are measured in all features. The number is the mean distance between two labs' centres, 6.2 within-plate spreads. Second view: one point per batch; agreement is the cosine between a batch's effect of 7 positive controls and its lab's first-batch effect; rho is Spearman after removing each lab's mean, minus 0.71 over 118 batches; shuffled, minus 0.03; without lab 2, minus 0.68; drift and effect from different wells, minus 0.72. Lab 2 against its June answer: 0.85 before the flag, 0.67 after.

**Precision to hold.** In lab 2 the flagged batches are also the last three, so inside that lab the flag and the calendar cannot be separated; the correlation uses every lab. The ringed points are one lab's autumn batches (A13); the crosses are batches whose positive controls show no effect (A8). If asked whether the map changes a lab's results: yes, modestly. In the field's frame, agreement between labs on the same drugs goes from 0.75 to 0.78, above the usual normalisation to a batch's own controls (0.76) in 107 of 125 batches; it does not change a lab's agreement with itself (A14).

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## 06 Open at the core (2:04–2:28)

*The standard. Point at the curve, then the three layers: open, paid, co-built. The moat and the business model in one breath.*

**2:04** Put them on one map and it sharpens with every lab that joins: six apart with one lab, under three with five. The map, the box and the code are open. The companies that buy cell data pay for it structured and comparable. And the benchmark is built with industry, so every method gets graded on it. *Click.*

**Precision to hold.** The prices are a proposal, not a validated model: about 5 to 15 thousand dollars per line per quarter for screening groups, a fee per certified lot. One unchecked screen is about 2.6 million, so the subscription is about 170 times less (A15). Bad-lot alerts across labs come after the certificate; don't promise them. If asked whether open-core works here: only if the free layer is never crippled to push the paid one.

**Figure detail.** Two references, each built from the same number of labs that share none; with one lab each they sit 6.2 within-plate spreads apart, with five each 2.8. The dashed line is one over root  $n$ . Eleven labs can make at most two disjoint sets of five, so the curve stops there. A lab's own baseline settles sooner: 0.89 with one plate, 0.39 with eight, then flat. The reference of all 11 labs moves 0.41 when one lab is removed.

**The three layers.** Community is open: the map, the box, the code, the public reference, labs by number, streaks and ranks. It has to stay fully usable for free, forever, or the standard dies. Data access is paid: the same data structured, comparable, pre-cleaned on request, with provenance and the certificate attached; the API, not the website; per certificate on bought data, per line per quarter for groups that screen continuously (A15). The benchmark is co-built: industry partners contribute failures and get their methods graded, segmentation to bioactivity. Whoever holds the benchmark holds the field.

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**07 Send your controls. See where you stand (2:28–3:04, two views)**

*What. The box first, then the map, then the two images at the bottom, then the answer card. Click: the leaderboard, and point at the pink rows.*

**2:28** The product is the box. Drop a control image; it stays yours.

**2:33** Back comes where your cells sit, whose they look like, their cells next to yours, and data every other lab can compare. Every drop sharpens the map, for you and for everyone. *Click.*

**2:48** And it's a game you can win. Every batch is ranked against the whole field on its controls alone: how typical, how tight, how steady. Your best, your streak, and the lab to beat. *Click.*

**Precision to hold.** This is a replay on public data, not a customer. The same lab's later batches sit outside its baseline from 30 August, and the certificate says so and says where to look: ER and AGP stain intensity (A11, A12, A13). Across the field 6 of 129 batches are outside. Bring that up if asked whether it ever says no. Brightfield is still the bet (A10).

**Figure detail.** One real lab on one real day: lab 2, 16 August, 25 plates, chosen as its in-distribution batch with the most plates. The image in the box is one of its control wells, DNA channel, as imaged; the second image at the bottom is a control well of lab 6. 2.8 spreads from its own baseline, 4.4 to lab 6, 188 cells per well. The reference is 1,652 plates from the other 10 labs. In the field's frame this batch's agreement with the other labs goes from 0.70 to 0.77. The ticks are the batches sent so far; the open one is the next.

**The game.** Field score is 100 minus the mean percentile of four measures on untreated wells alone: distance from the field's centre, spread of the batch's plates, distance from the lab's previous batch, and cells per well against the field's median. Lab 5 holds 9 of the top ten and a streak of 33. Lab 2's best is #48 of 129; the batch on the slide is #95. Nothing in the score touches a treated well, so the only way to climb is better culture and imaging practice; say that if anyone asks whether a leaderboard is bad for science (A17).

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## 08 Five questions, five tools (3:04–3:22)

*Point at the open tick in the Datum Line column: identity is done with a partner, STR. One sentence.*

**3:04** Five tools, five questions. STR: which line. The vendor: fine when shipped. Plate controls: the plate worked. Platforms: your own data, consistent. None say where your cells sit against every other lab. Use all of them; the last one didn't exist. *Click.*

**Precision to hold.** Say what each tool does well before what it does not. Datum Line's own gaps are on the slide: it cannot tell which line it is, and it is fluorescence only today. STR is the partner for identity (A7). Phenomics platforms are not named; if asked, the point is that batch correction inside one organisation removes the lab signature on purpose, which is right for discovery and wrong for a reference (A3).

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## 09 Eighteen months, four milestones (3:22–3:34)

*The money. Four cards left to right; the pink one is the milestone that pays.*

**3:22** Eighteen months: three labs live on the map, brightfield proven, the benchmark built with two industry partners, and a first paid certificate on data somebody bought. *Click.*

**Figure detail.** Milestones, not dates. The pre-seed is about \$1M for eighteen months, built line by line on A21 and every line a proposal: people first, the benchmark inside it, no lab of our own. Three labs on the map means the collaborating labs, insitro, Axiom Bio and Soley, sending U2OS controls every batch and reading answers; say "collaborating". Brightfield proven means the same map from label-free images (A10). The test set is the ugliest dataset on purpose: six labs' real wells plus planted failures, bubble, focus, exposure, clipped well, bad lot, wrong label, with every method graded on naming the failure (A16). A first paying line is one pharma cell line on a per-quarter subscription.

**Why the ugly dataset.** Because whoever holds the benchmark holds the field. Six labs' real wells plus planted failures, public, with every method graded on the same thing: does it name the failure, does it normalise without erasing the lab, does it recover the bioactivity that was put in. Every normalization paper, every foundation model, every CRO's QC gets measured against it, so it becomes the glue between the labs that make data and the companies that buy it: both sides grade against the same set. The cheapest way to own a standard is to build the test everyone has to pass (A16).

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## **10 Every plate, placed on the map (3:34–3:59)**

*Vision, then the ask. Point at the open row, then the right card. Stop on the question.*

**3:34** Every plate you image, placed on the map. Already there, from public data: 129 batches, eleven labs. The next row is yours. People read this map; agents train on it. Everyone learns and grows together.

**3:48** What I need: images, U2OS first; a technical co-founder; bench time; a pre-seed; and two partners for the benchmark.

**3:56** Have images of U2OS cells? Talk to me.

**Precision to hold.** Price and test-set budget are not set: say "per line, per quarter" and "the proof of concept sets the size". Don't name a partner lab until one has agreed. Name no JUMP lab beyond its number. The grid is 129 batches and 1,871 plates; each square is a batch. Now is U2OS in fluorescence; more lines and brightfield are next and are not yet shown.

**Why here.** San Francisco has the densest cluster anywhere of companies that buy cell data they did not make and train models on it, plus JUMP partners. The first ten customers are a walk away, and two of them are on the resume. Name none of them as customers until one has agreed; say "the companies that buy cell data".

The technology, if asked. Today it is geometry, not a model: 3,773 CellProfiler numbers per well, z-scored on the other labs, one centroid per plate, a logistic regression to name the lab, a distance to place a batch, one matrix per batch to move it into the field's frame. The whole replay runs on a laptop from cache in minutes. Next is any image: a frozen vision model placing a raw picture (results/e15), then a model of cells trained on the open map and graded on the benchmark. Treated data never moves: the frame goes to the lab, and later, models learn from treated data where it sits. Align first; only then can learning cross labs. People use the box and the game; agents use the API and the benchmark; both learn from the same map (A19).